



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
- M07: LLM Fundamentals
- M08: Prompt Engineering
- M09: Fine-Tuning & PEFT
- M10: RAG & Real-World Projects



Advanced NLP & Large Language Models (LLMs)

4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

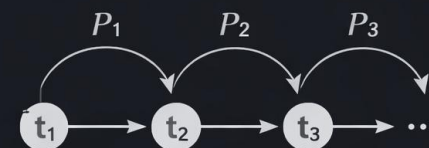
4.6 Summary and The Entity Lifecycle

M04 Entity-Centric NLP

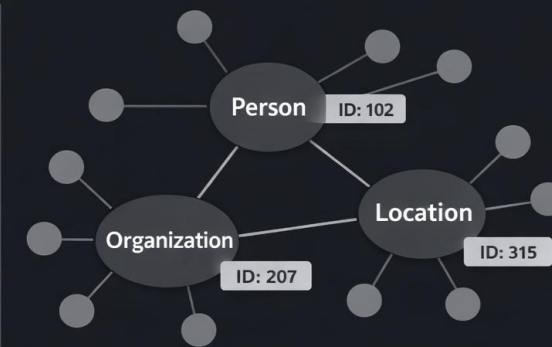
Chapter 4.1 – Introduction to Entity-Centric NLP

From Sequence Modeling to Entity-Centric Intelligence

- Traditional NLP models language as token sequences and probabilities
- High-reliability AI requires stable, identifiable real-world objects
- Entity-Centric NLP shifts from linguistic tokens to knowledge objects
- Entities enable tracking, memory, and reasoning across documents
- This marks the transition from fluent language to grounded intelligence



Sequence Modeling



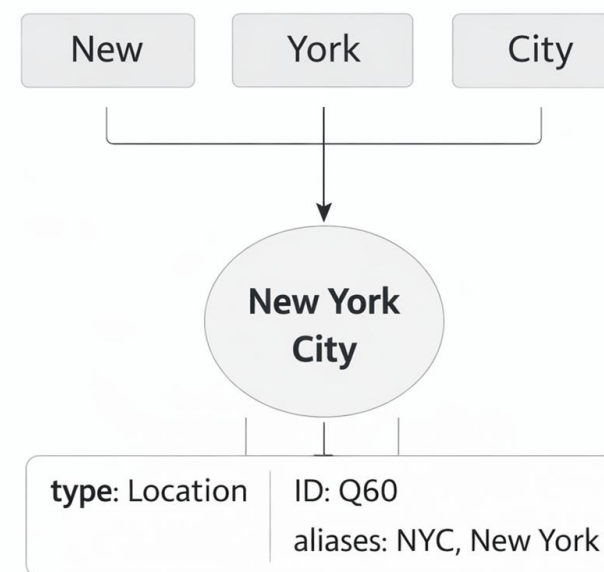
Entity-Centric Intelligence

M04 Entity-Centric NLP

Chapter 4.1 – Introduction to Entity-Centric NLP

What Is an Entity? The Atom of Knowledge

- An entity is a unique, persistent real-world or domain-specific object
- Unlike tokens, entities maintain identity across contexts and documents
- Examples include people, locations, organizations, laws, or chemicals
- Entities are the stable units stored in knowledge graphs and databases
- They enable long-term memory and cross-document reasoning





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Chapter 4.2 – NER: Identification and Classification (Index)

What Is Named Entity Recognition?

- NER is a sequence tagging task over tokenized text
- Each token is assigned a label indicating entity membership
- Labels encode both boundaries and semantic categories
- The output transforms raw text into structured entity spans
- NER provides the foundation for all downstream entity tasks



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Chapter 4.2 – NER: Identification and Classification (Index)

Standard Entity Types and Taxonomies

- NER systems classify entities into predefined semantic categories
- Core universal types form the base of most industrial systems
- Extended taxonomies adapt NER to specific domains
- Correct typing is critical for downstream reasoning and linking

Core entity categories

- PER – People and individuals
- ORG – Organizations and institutions
- LOC – Locations (non-political)
- GPE – Countries, cities, geopolitical regions

Extended / domain types

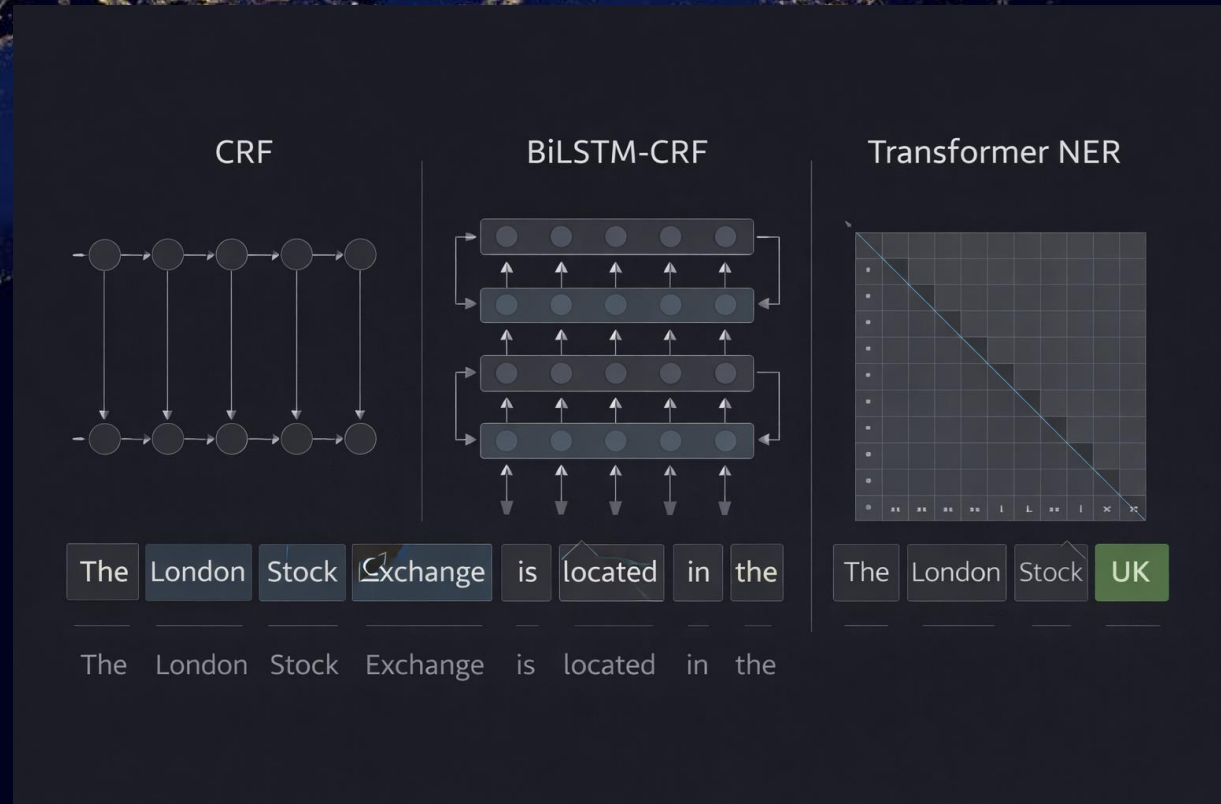
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Chapter 4.2 – NER: Identification and Classification (Index)

Sequence Tagging Approaches

- NER is framed as a sequence labeling problem
- Models predict a label for each token in context
- Approaches evolved from probabilistic to deep neural architectures

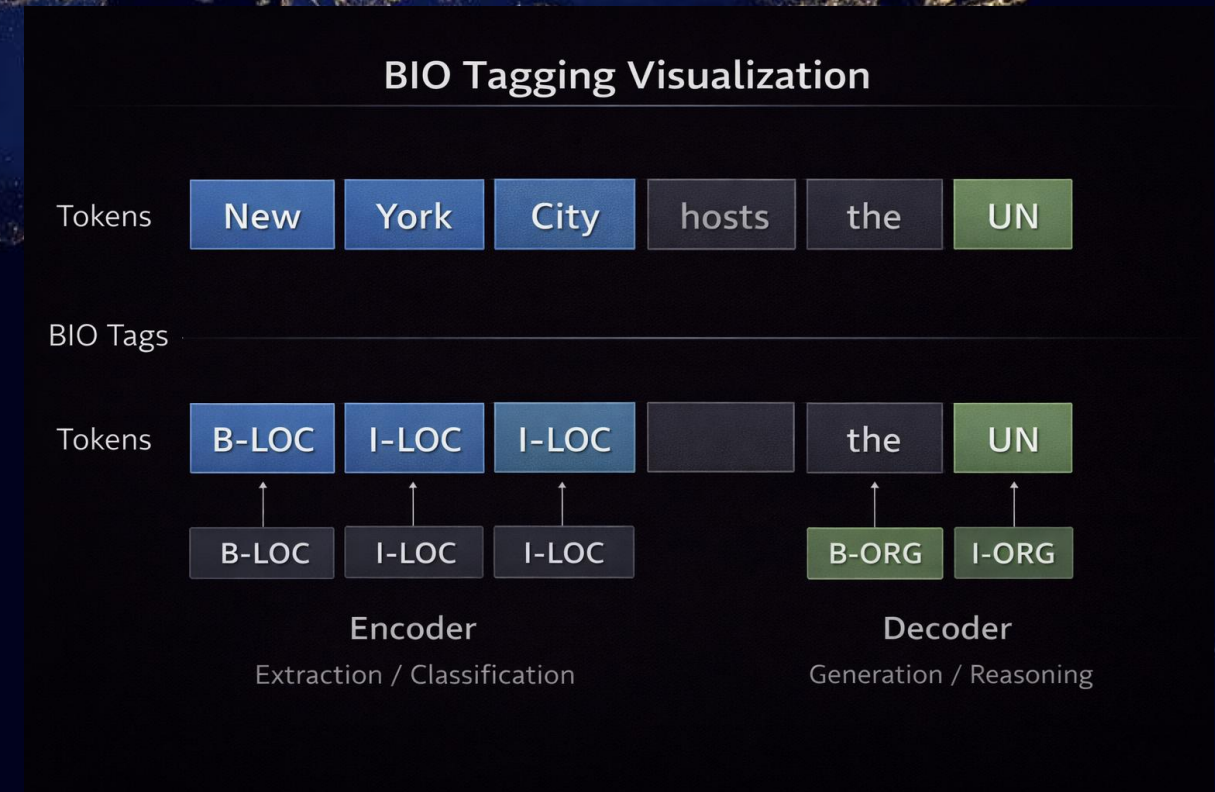


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Chapter 4.2 – NER: Identification and Classification (Index)

Annotation Schemes: BIO / IOB Tagging

- Multi-token entities require boundary-aware labeling
- BIO (or IOB) is the most widely used annotation scheme
- Each token receives a boundary + type label



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Chapter 4.2 – NER: Identification and Classification (Index)

Evaluation Metrics for NER

- NER evaluation is span-based, not token-based
- Correct prediction requires exact boundary + label match
- Partial matches count as errors

NER Evaluation

Gold entity spans

Predicted entity spans

True Positive

London Stock Exchange = Located in

False Positive

London Stock Market ≠

False Negative

UK ≠ US

The London Stock Exchange is located in the UK

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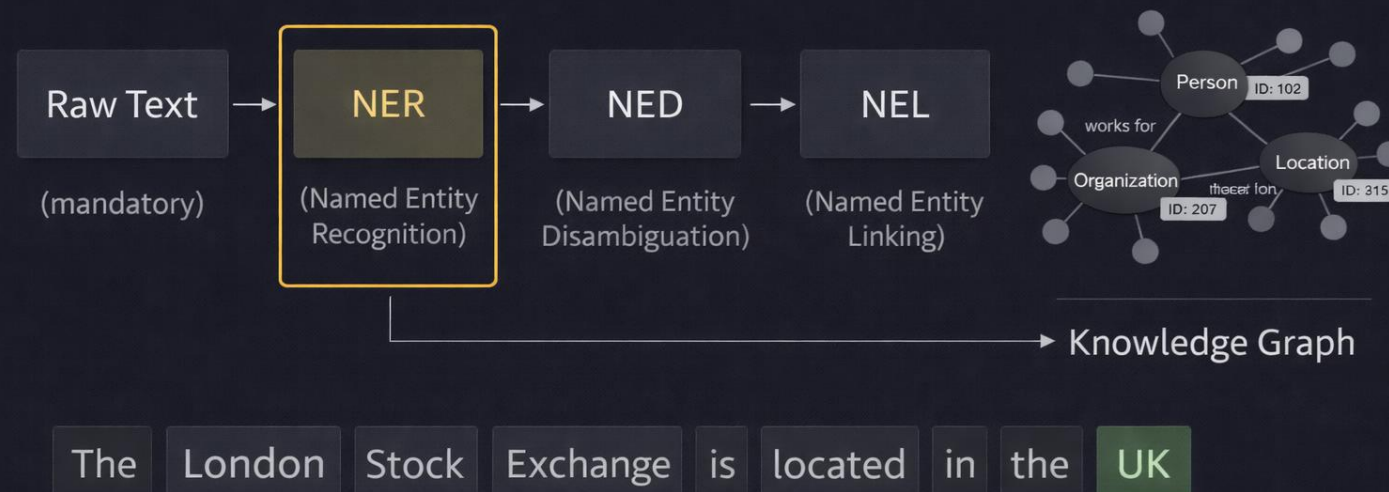
Chapter 4.2 – NER: Identification and Classification (Index)

Why NER Is the Foundation of Entity-Centric NLP

- NER is the entry point to entity intelligence
- All downstream tasks depend on accurate detection
- Errors propagate if NER fails

NER → NED → NEL → Relation
Extraction → Knowledge Graphs

Entity-Centric NLP Pipeline





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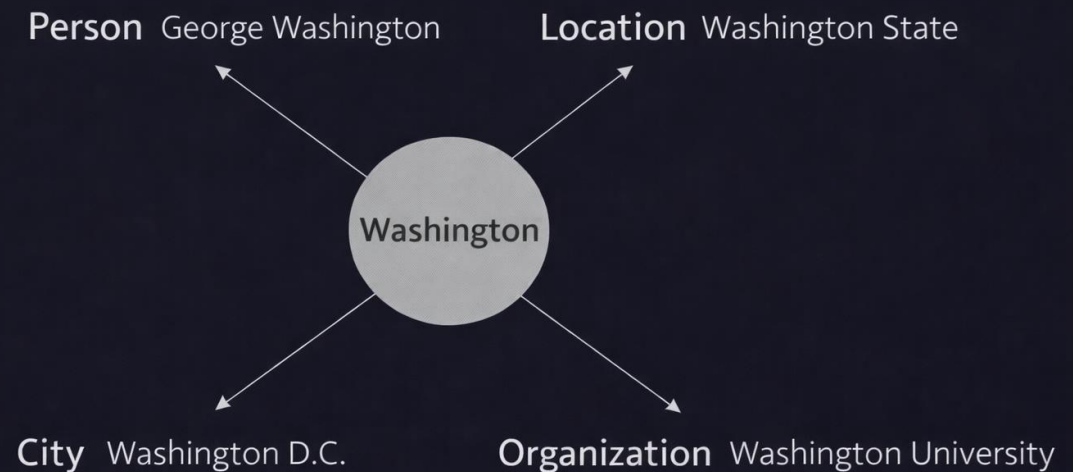
4.6 Summary and The Entity Lifecycle

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Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

The Problem of Entity Ambiguity

- The same surface name can refer to multiple real entities
- Ambiguity is unavoidable in natural language
- NER alone cannot determine the correct referent
- Unresolved ambiguity leads to factual errors

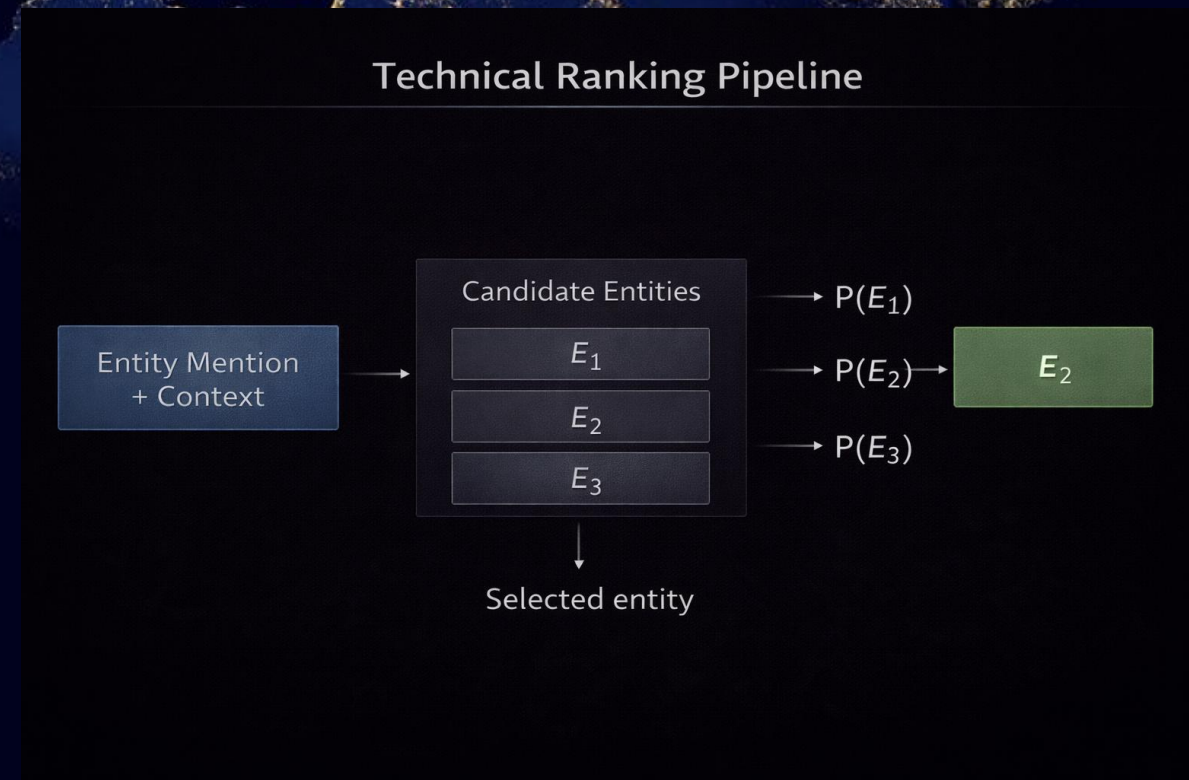


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Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

What Is Named Entity Disambiguation (NED)

- NED selects the correct entity among candidates
- It is modeled as a ranking problem
- Input: mention + context
- Output: single most probable real-world entity

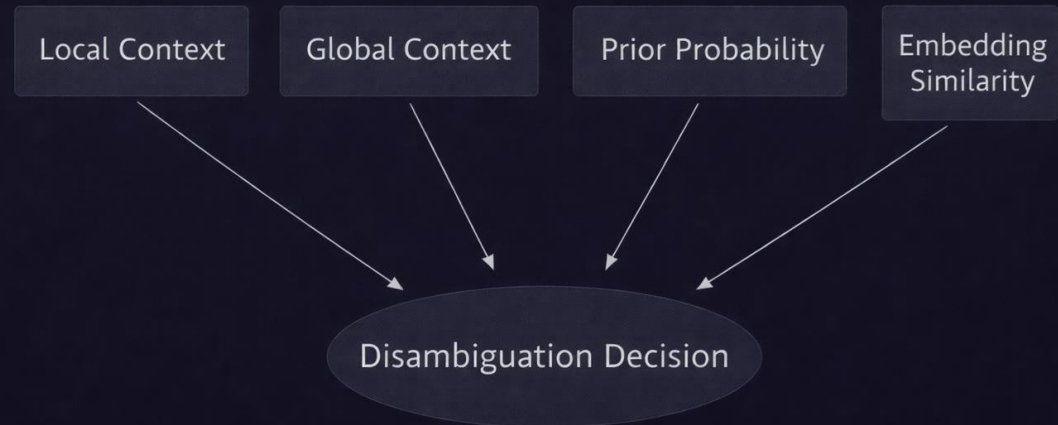


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Signals Used in Entity Disambiguation

- Local context around the mention
- Global document coherence
- Prior probability from knowledge bases
- Semantic similarity in embedding space

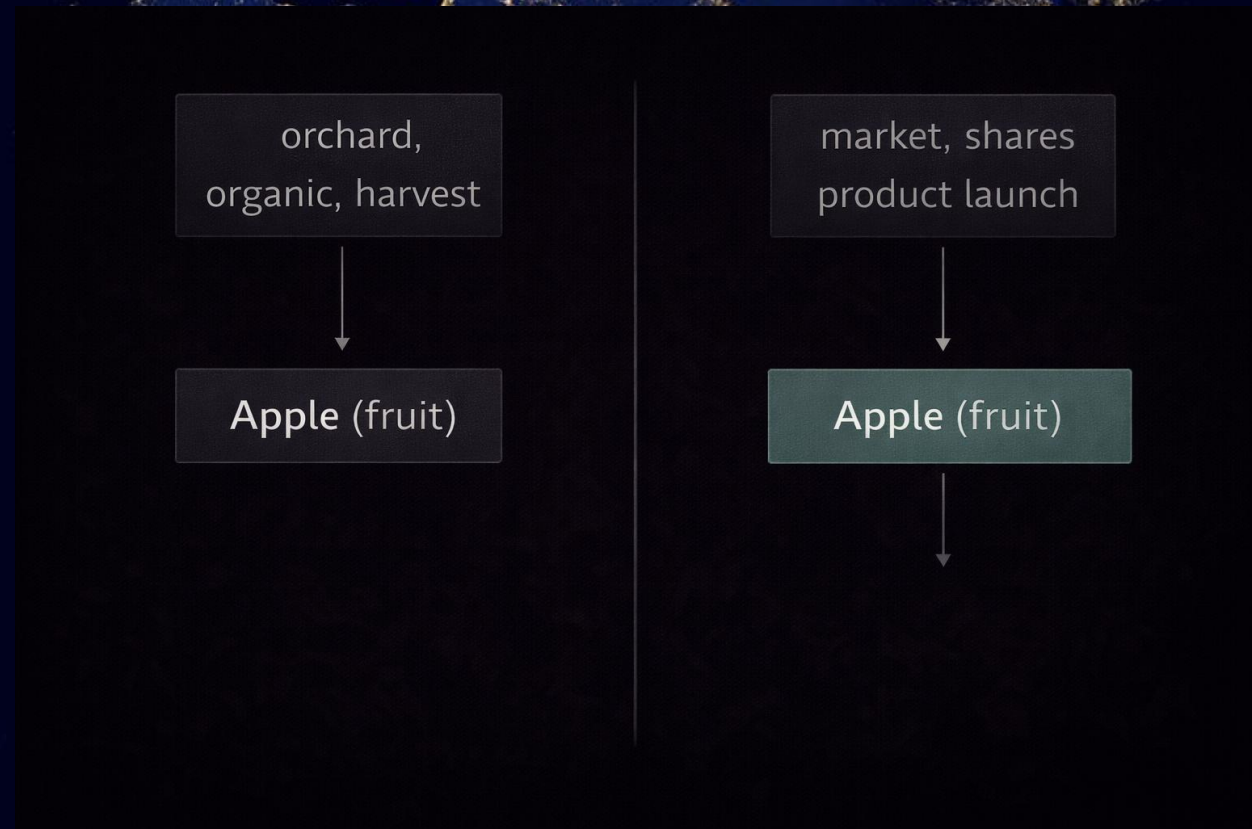


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Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Example: Resolving Ambiguous Mentions

- Context determines the entity meaning
- Semantic cues guide the resolution
- Same word, different real-world entities
- Correct resolution enables safe reasoning

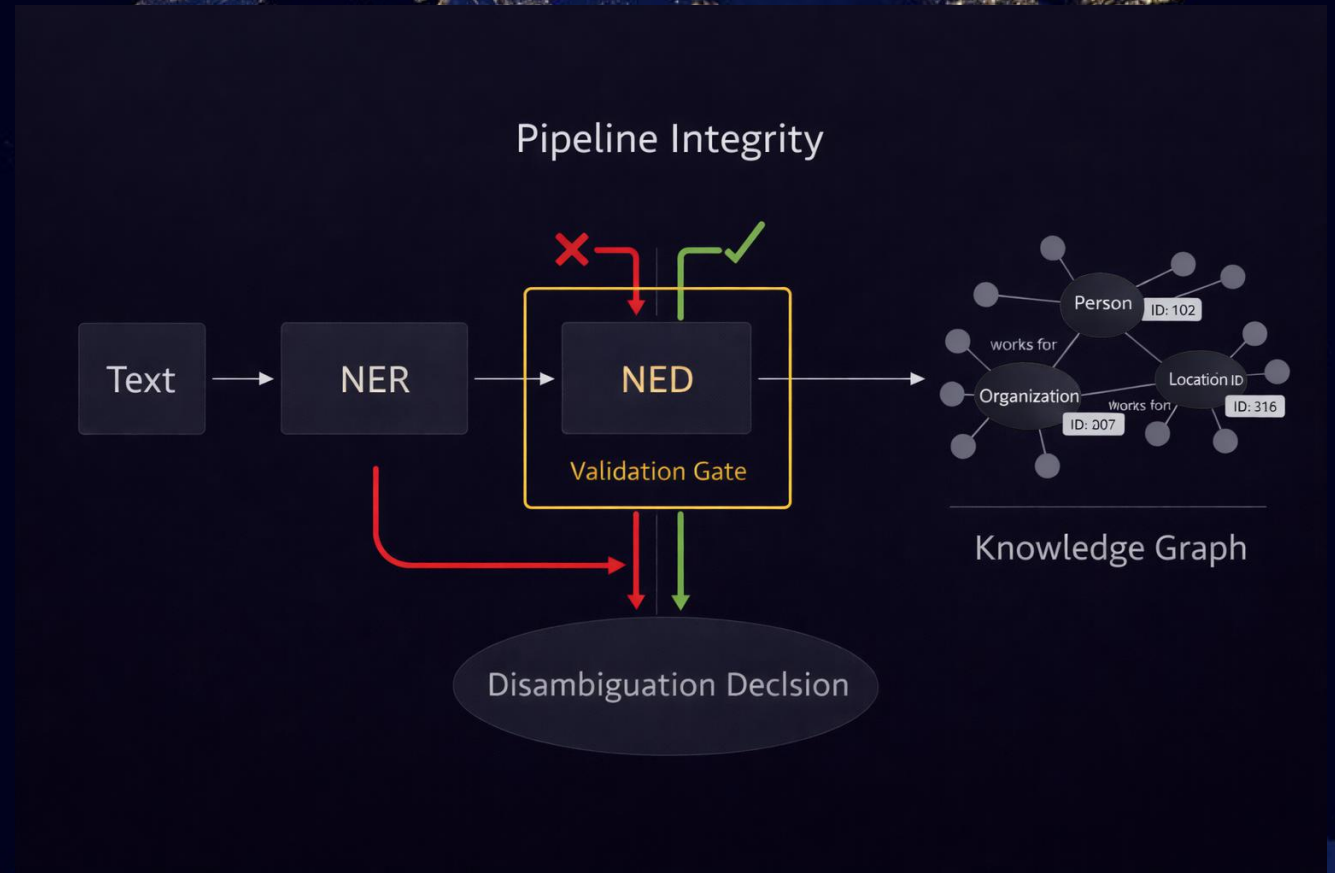


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Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Why NED Is Critical for Knowledge Systems

- Prevents incorrect facts entering knowledge graphs
- Enables precise analytics and reasoning
- Required before entity linking (NEL)
- Core reliability layer in enterprise AI





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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

What Is Named Entity Linking (NEL)

- NEL connects text mentions to real-world entities
- Each entity is resolved to a unique identifier
- Eliminates ambiguity across aliases and synonyms
- Enables persistent, machine-readable references

Conceptual Grounding

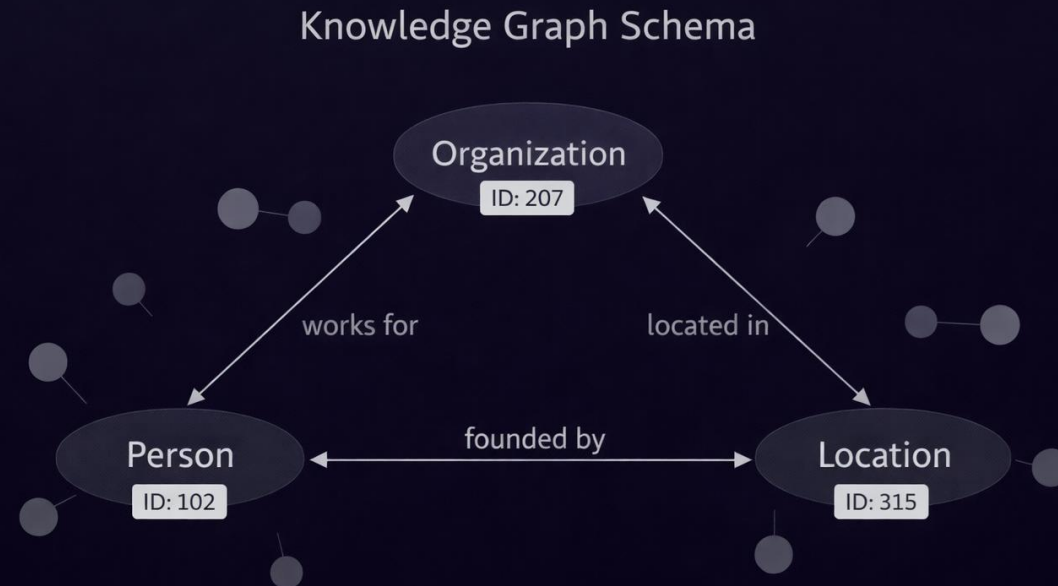


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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Knowledge Bases and Unique Identifiers

- Knowledge Bases store structured world facts
- Entities are identified by global IDs
- Relations are stored as subject–predicate–object triples
- Enables reasoning beyond raw text



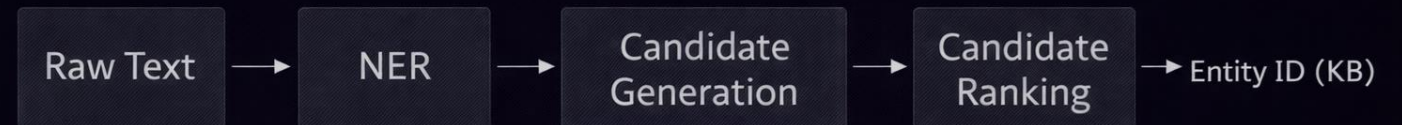
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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

The NEL Engineering Pipeline

- Detect entity mentions (NER)
- Generate candidate entities
- Rank candidates using context (NED)
- Link to knowledge base ID
- Handle unknown entities (NIL)

End-to-End NEL Pipeline



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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NED vs NEL: Key Differences

- NED selects the correct candidate
- NEL implements the full linking system
- NED is a decision task
- NEL is an end-to-end architecture

NED vs. NEL: Key Differences

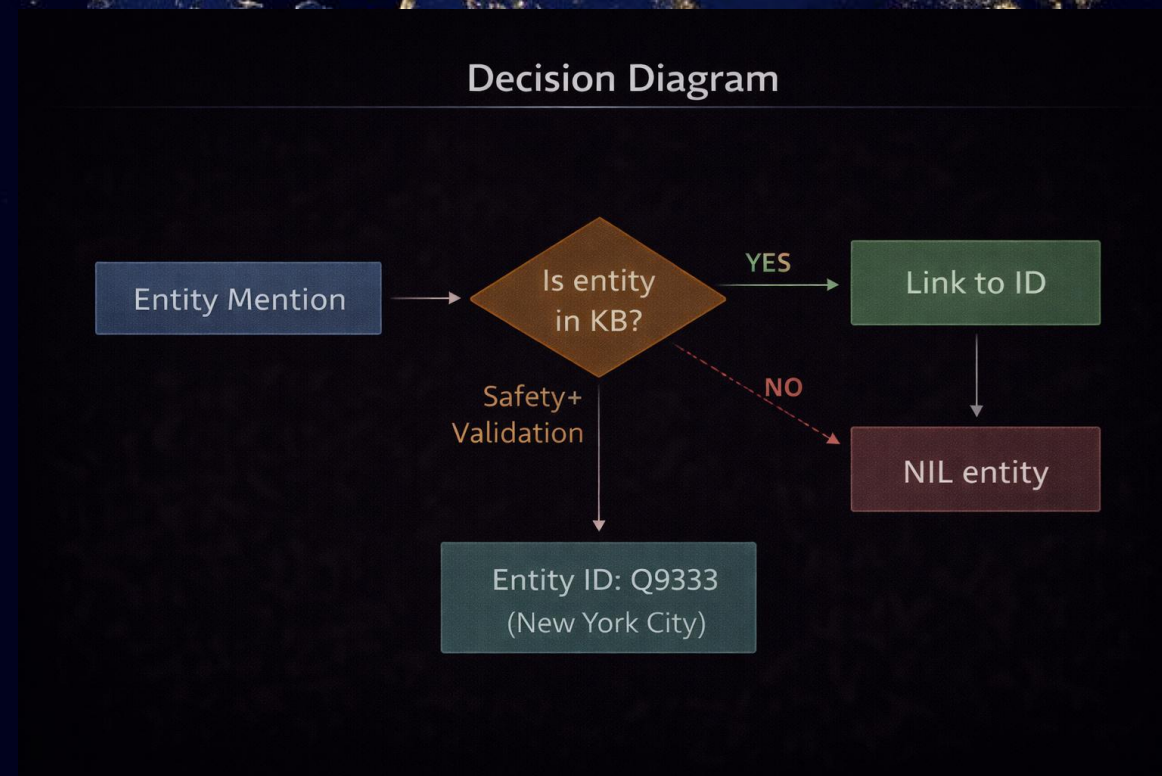
NED (Named Entity Disambiguation)	NEL (Named Entity Linking)
Scope Focuses on identifying which known entity an ambiguous mention refers to WITHIN a document/context	- Connects mentions of entities in entries. gl. Wikipedia, Widarat) - Connects mentions of entities in text text to entries in knowledge base (e. gl. Wikipedia, Widarat)
Output A specific entity ID or reference from predefined set	A URI/URL or unique identifier to a knowledge base entry
Complexity Context-dependent; requires local analysis	Higher Requires robust knowledge base lookup & entity resolution
Role in pipeline Often a sub-component of a larger NLP system	Often a downstream task from NED; enriches text with external knowledge

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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NIL Detection and Real-World Constraints

- Not all entities exist in knowledge bases
- Systems must detect NIL cases safely
- Prevents hallucinated or false entities
- Critical for private or emerging entities

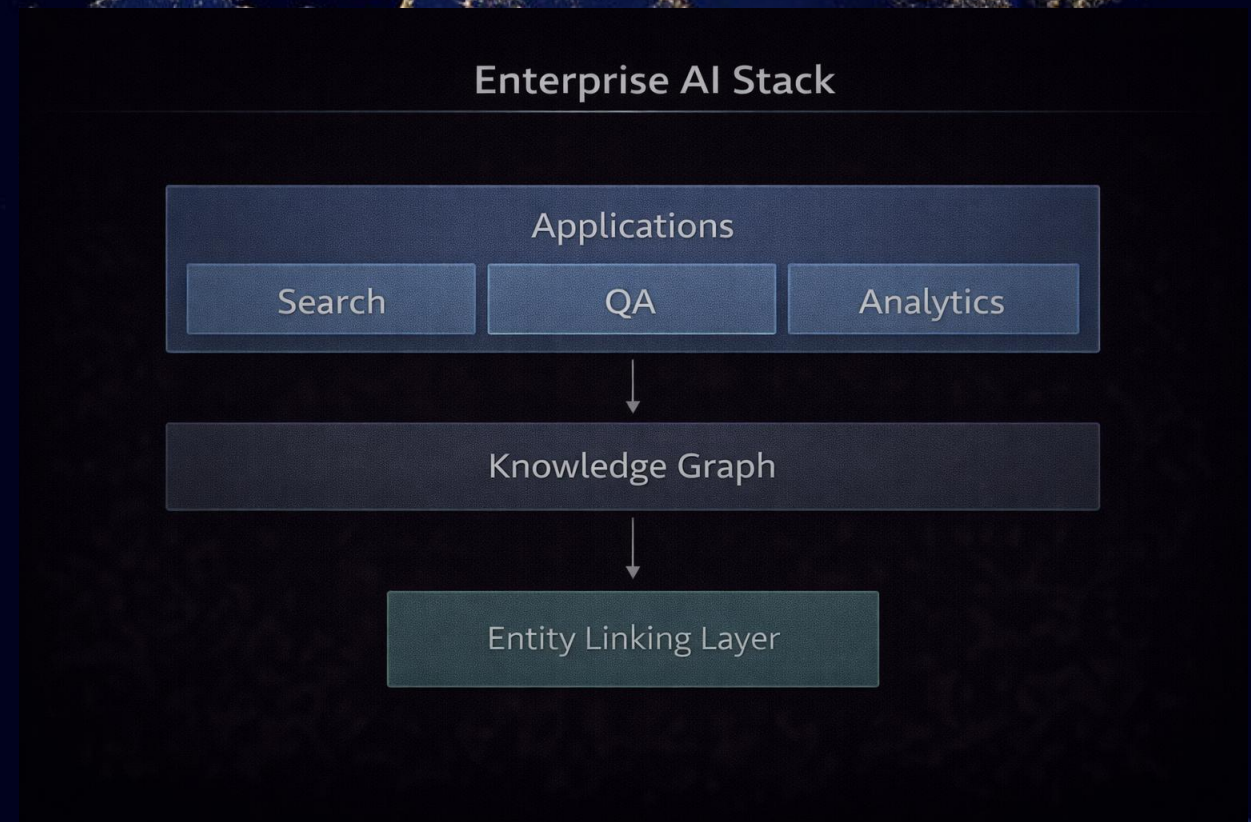


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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Why NEL Is Essential for Reliable AI

- Grounds language in factual reality
- Enables explainable reasoning
- Supports analytics and traceability
- Foundation for enterprise knowledge graphs





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Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Entities Matter

- General NER fails in specialized domains
- Domain entities are longer and more complex
- Labels are highly specific and non-standard
- Precision is critical in regulated environments

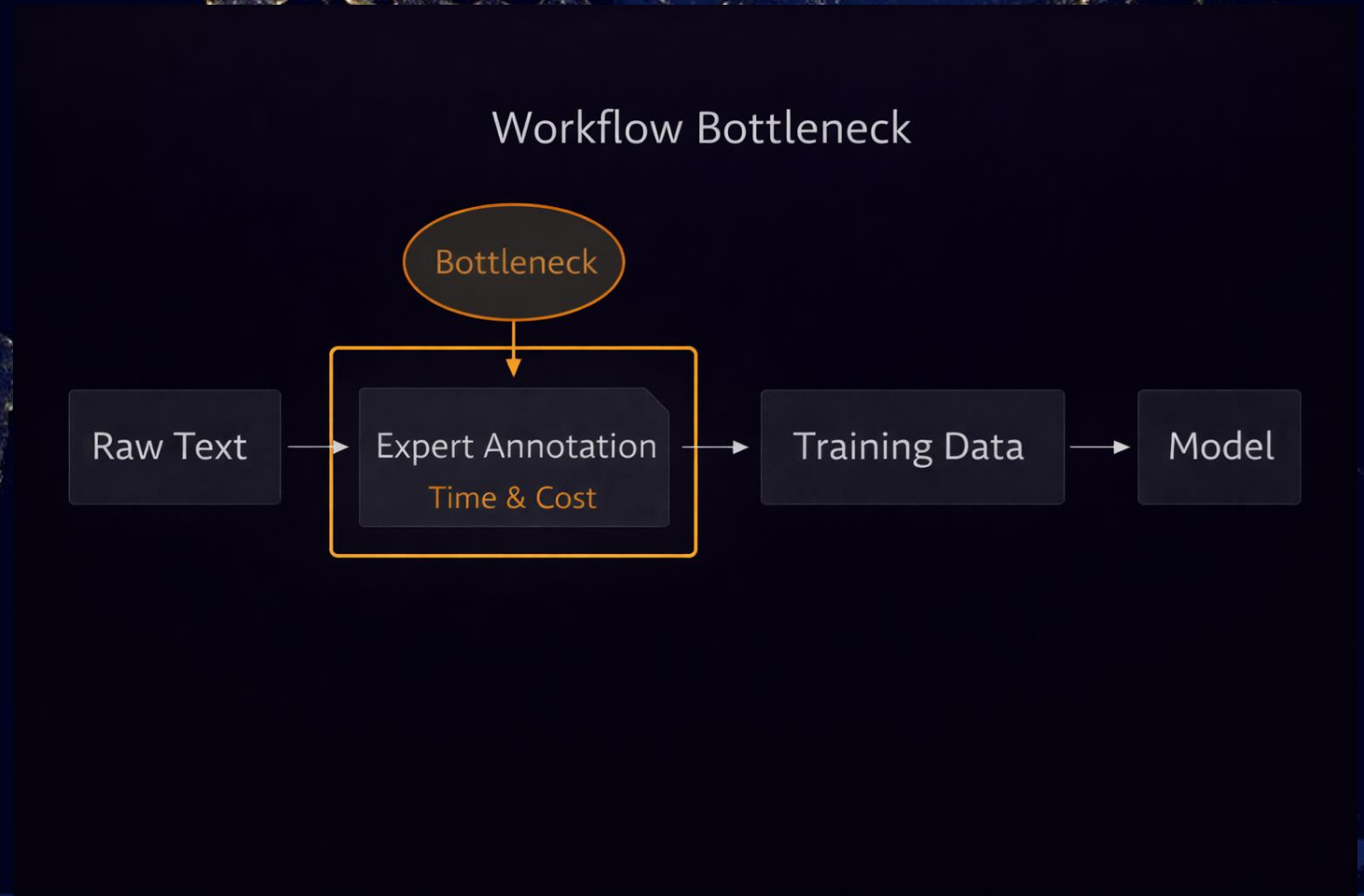
General NER	Domain NER
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition

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Chapter 4.5 Advanced Domain-Specific Entity Extraction

The Annotation Bottleneck

- Domain annotation requires expert knowledge
- Manual labeling is slow and expensive
- Data scarcity limits supervised training
- Annotation quality impacts downstream accuracy

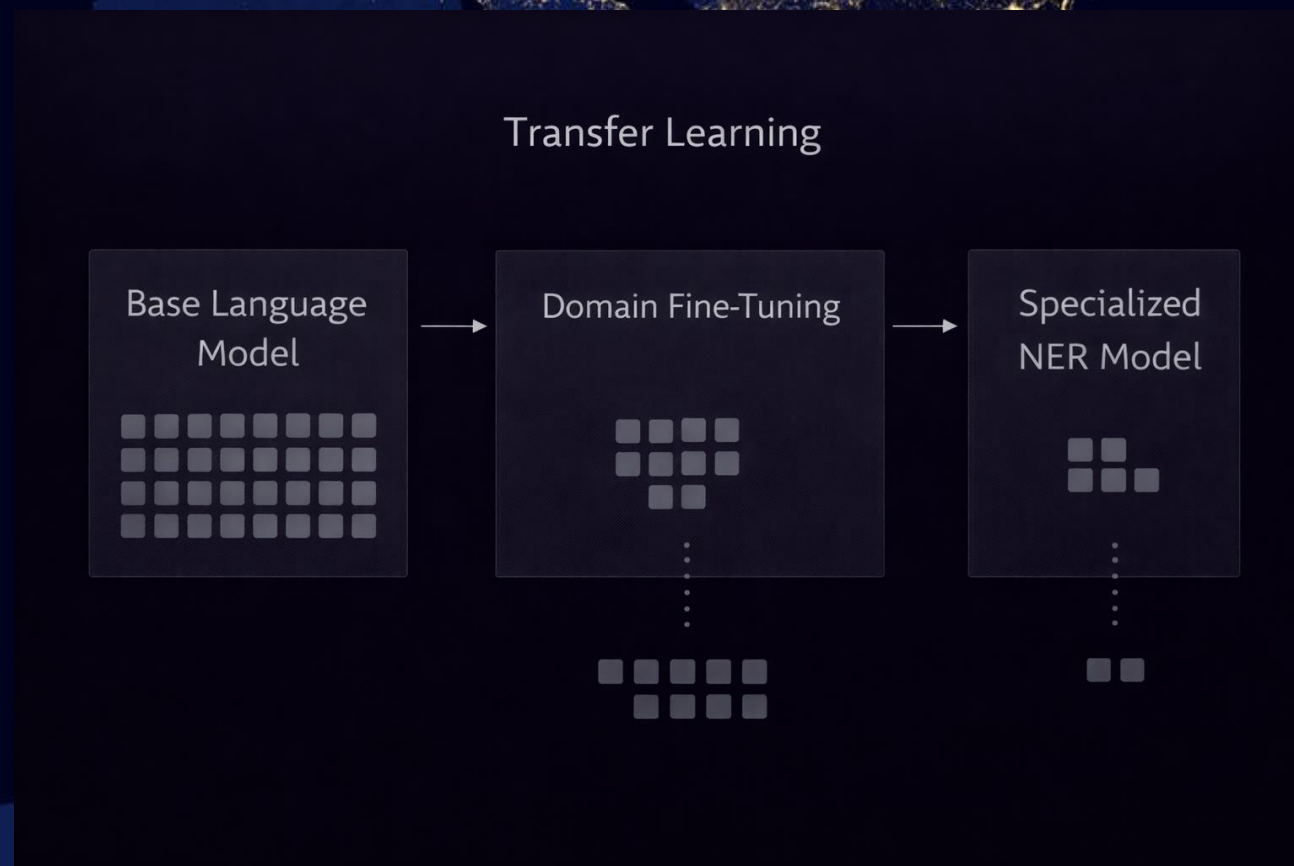


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Transfer Learning for Specialized NER

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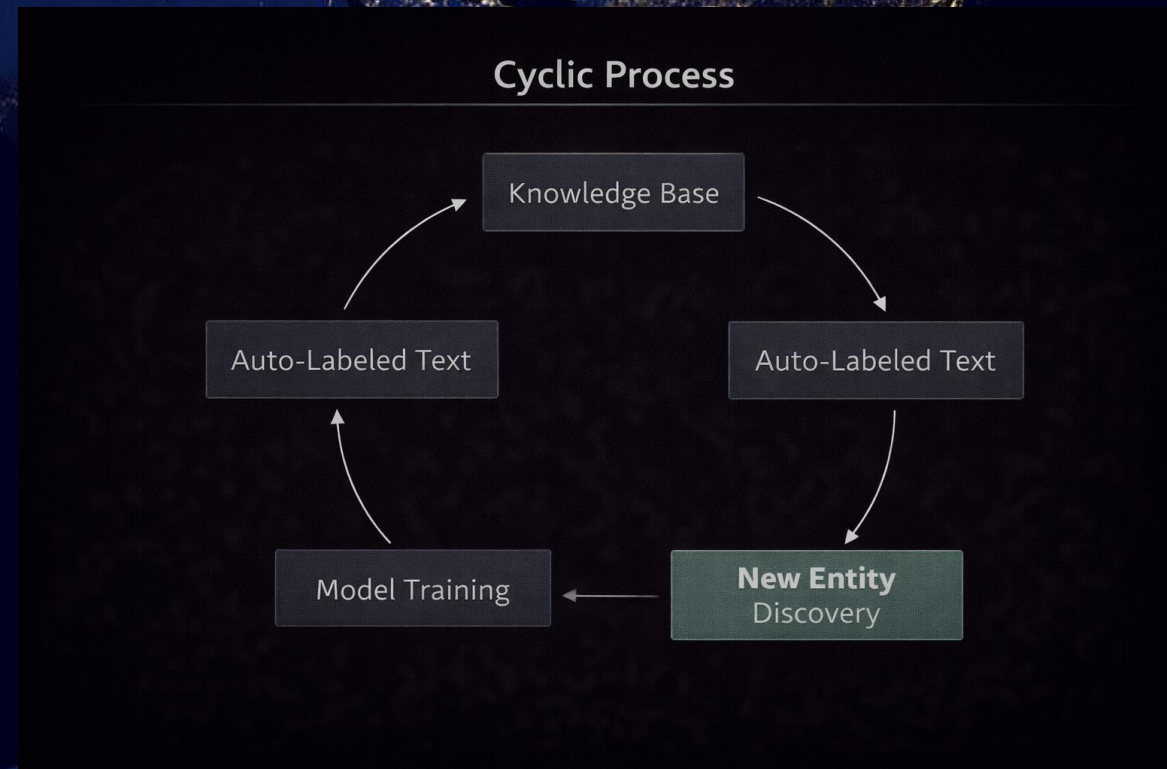


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Distant Supervision and Bootstrapping

- Use existing databases as weak labels
- Automatically annotate large corpora
- Iteratively expand entity coverage
- Produces scalable “silver” datasets



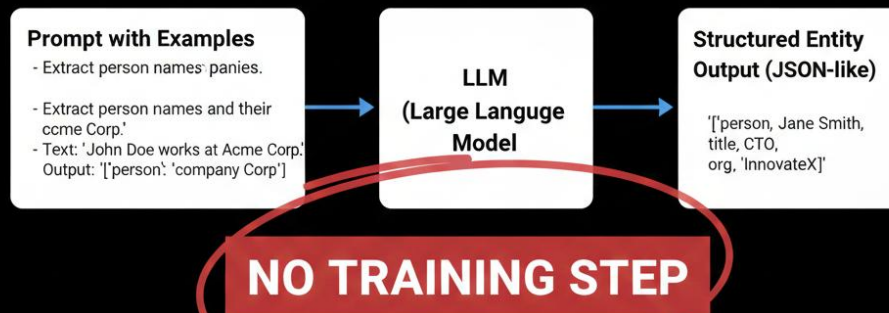
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Chapter 4.5 Advanced Domain-Specific Entity Extraction

LLM Prompting for Entity Extraction

- Zero-shot extraction via instructions
- Few-shot examples improve precision
- Rapid adaptation without retraining
- Outputs used as silver labels

Prompt-Based Information Extraction



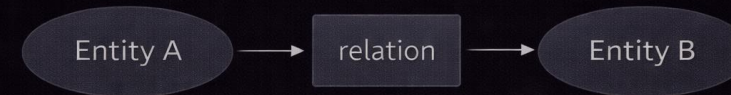
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Chapter 4.5 Advanced Domain-Specific Entity Extraction

Relation Extraction as the Next Step

- Entities alone are not sufficient
- Relations capture semantic structure
- Output expressed as triples
- Enables dynamic knowledge graphs

Knowledge Triples



Examples:

Drug → treats → Disease

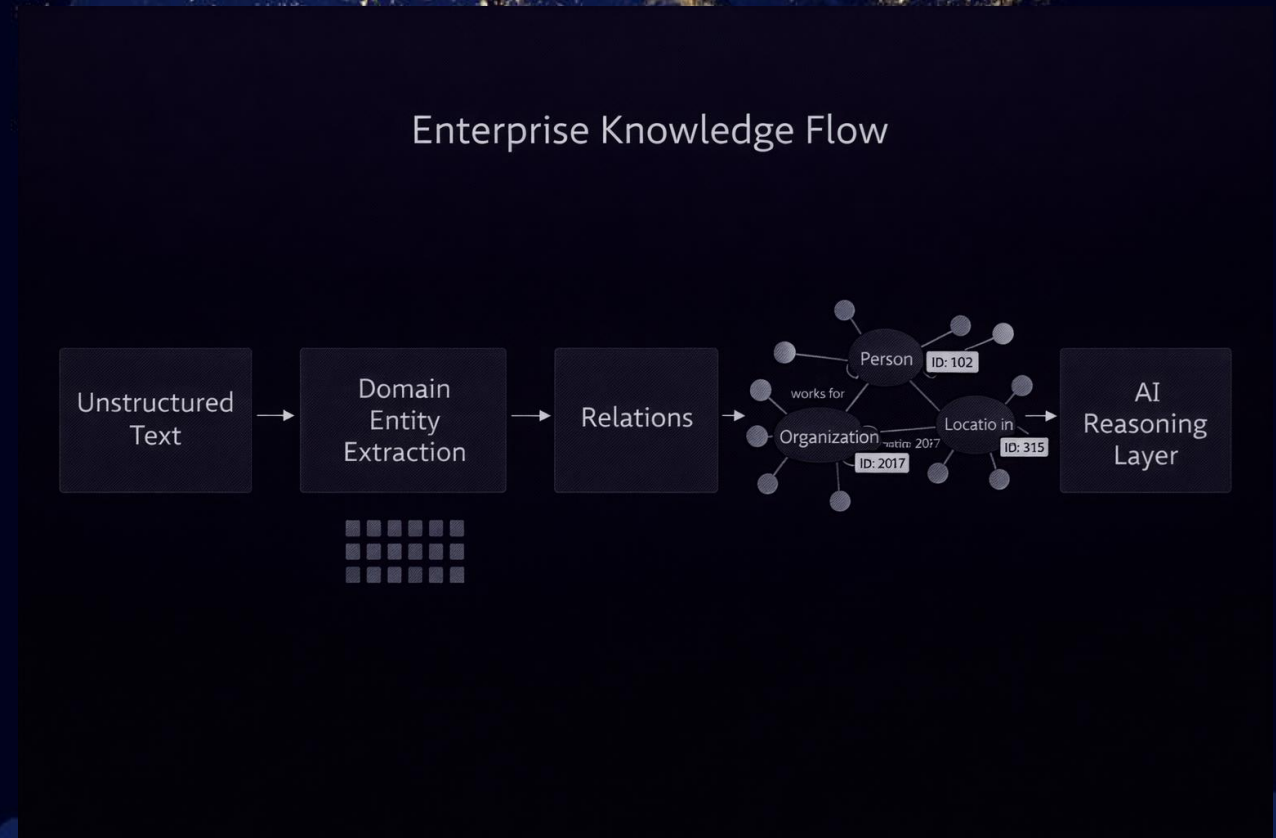
Company → headquartered in → City

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Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Extraction Enables Reasoning

- Transforms documents into structured data
- Supports automated reasoning systems
- Enables real-time knowledge updates
- Foundation for decision intelligence





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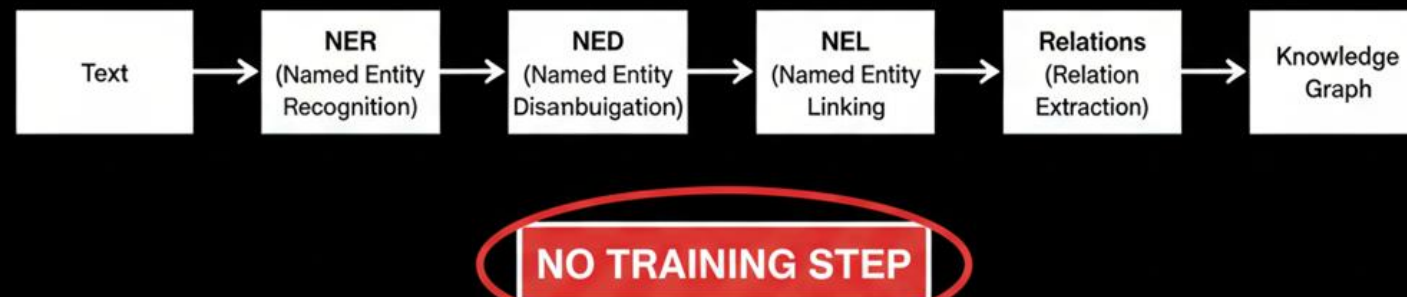
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Chapter 4.6 Summary and The Entity Lifecycle

Recap of the Entity-Centric Pipeline

- Entities are the atomic units of meaning
- NER identifies entity spans
- NED resolves ambiguity
- NEL grounds entities in knowledge bases
- Relations connect entities semantically

Linear NLP Recap Pipeline

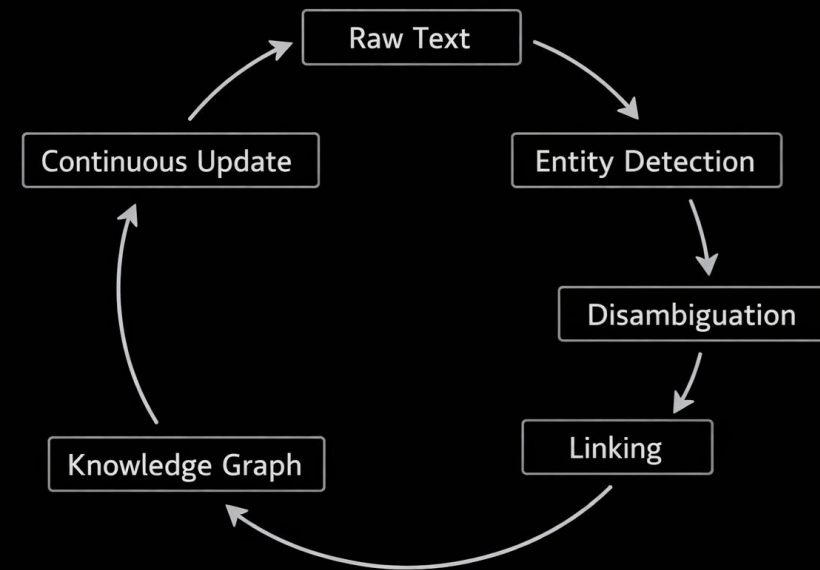


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Chapter 4.6 Summary and The Entity Lifecycle

The Complete Entity Lifecycle

- Entities emerge from raw text
- Entities are validated and linked
- Entities evolve over time
- Knowledge graphs are continuously updated



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Chapter 4.6 Summary and The Entity Lifecycle

Key Challenges in Real-World Entity Systems

- Low-resource and emerging entities
- Ambiguous or overlapping mentions
- Knowledge base incompleteness
- Entity drift over time

CHALLENGE MATRIX



Low-resource entities

Limited information availability



Ambiguity

Contextual uncertainty



KB gaps

Missing knowledge in knowledge bases



Temporal drift

Changes over time

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Chapter 4.6 Summary and The Entity Lifecycle

Role of Generative Models in Entity NLP

- LLMs act as internal annotators
- Enable zero-shot entity extraction
- Assist relation discovery
- Must be validated to prevent hallucinations



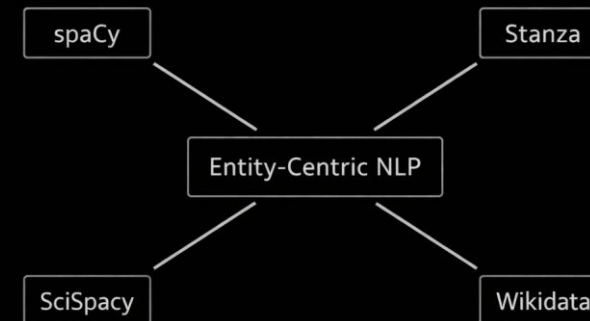
Hybrid AI architecture

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Practical Tools and Libraries

- spaCy: production-grade NER pipelines
- Stanza: multilingual neural NLP
- SciSpacy: biomedical entity extraction
- Wikidata: global knowledge grounding



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Chapter 4.6 Summary and The Entity Lifecycle

Final Takeaways

- Entities anchor language to reality
- Entity grounding reduces hallucinations
- Knowledge graphs enable explainability
- Entity-centric NLP is essential for trusted AI

CONCEPTUAL TAKEAWAY DIAGRAM



Text probabilistic layer →



Entity grounding layer →



Reliable AI system

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Chapter 4.6 Summary and The Entity Lifecycle

Transition to the Next Module

- Entity-centric NLP enables downstream tasks
- Grounded entities improve analytics accuracy
- Next: applying entities to business problems
- Sentiment, QA, and summarization

TRANSITION DIAGRAM



Entity-Centric NLP



Business NLP Tasks:

- Sentiment Analysis
- Question Answering
- Summarization

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M04 – Academic References and Research Resources

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